EXP NO: 14. PACKET SNIFFING USING RAW SOCKETS

Aim:

To implement packet sniffing using raw sockets.

code:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP
def packet_callback(packet):
  if IP in packet[IP]
  protocol = ip_layer.proto
  src_ip = ip_layer.src
  dst_ip = ip_layer.dst
  protocol_name = ""
  if protocol_name == 1:
    protocol_name = "ICMP"
  elif protocol_name == 6:
    protocol_name = "TCP"
  elif protocol_name == 17:
    protocol_name = "UDP"
  else:
    protocol_name = "unknown protocol"
  print(f"protocol: {protocol_name}")
  print(f"source IP: {src_ip}")
  print(f"Destination IP: {dst_ip}")
  print("_" * 50)
sniff(iface = 'Wi-Fi', prn = packet_callback, filter="ip"
      store = 0).
```

Output:

```
Protocol: UDP
Source IP: 57.144.53.54
Destination IP: 192.16.57.165
```

protocol : UDP
Source IP : 57.144.53.54
Destination IP : 172.16.57.165

protocol : UDP
Source IP : 57.144.53.59
Destination IP : 172.16.57.165

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